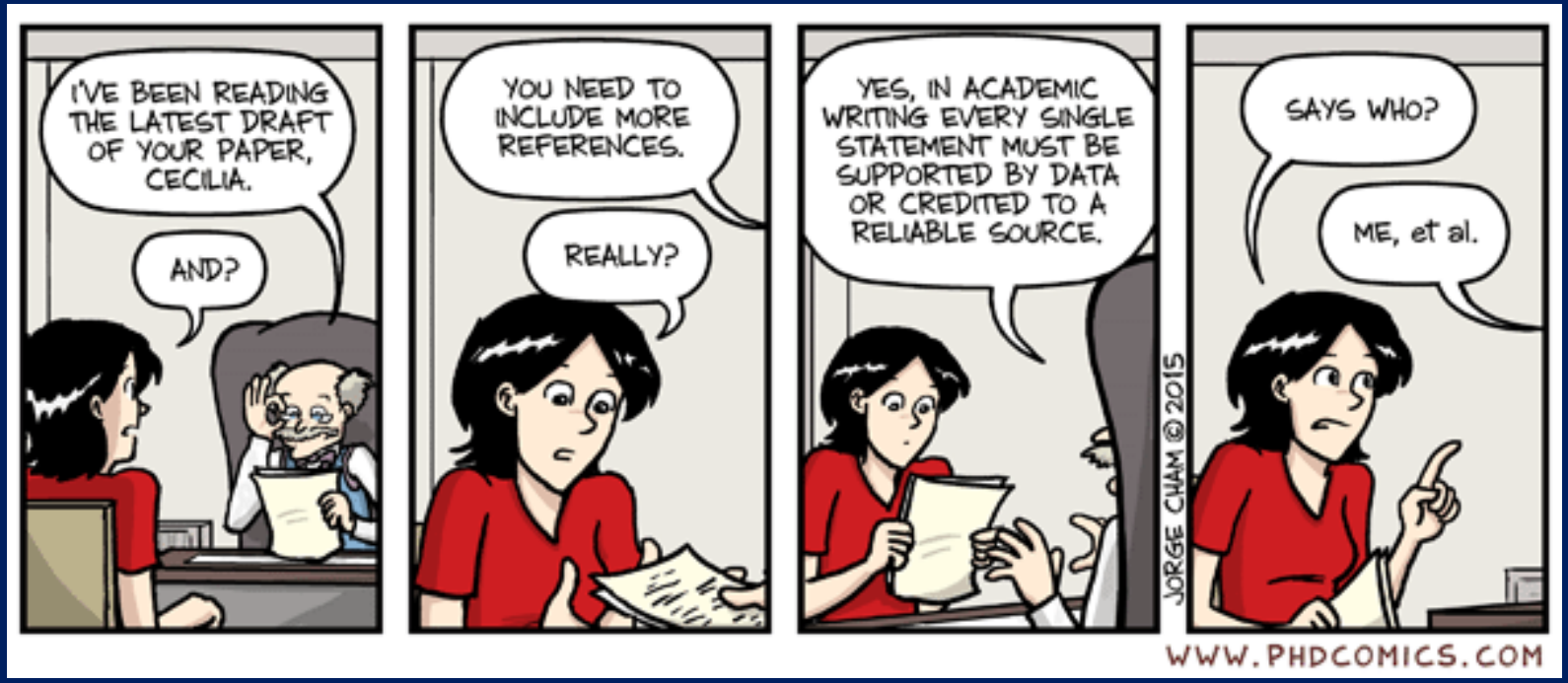
Research Skills Class 2:

How to find and use references



What are references??

Scientific references are cited sources:

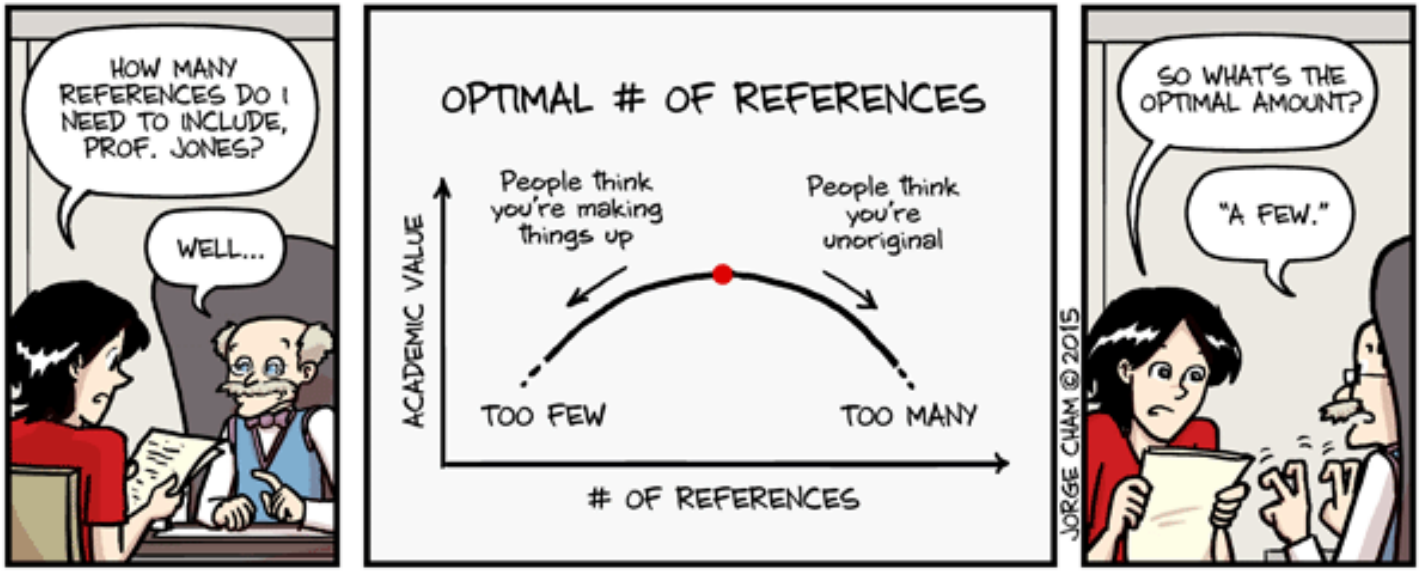
1. Showing where you obtained your data or information
2. That readers can consult for more details on a topic

1. INTRODUCTION

The joint analysis of multiple cosmological observables is a key approach for defining and ultimately resolving the outstanding questions of cosmology (for recent reviews see, e.g., Weinberg et al. 2013; Moresco et al. 2022; Huterer 2023). These puzzles include the approximately 5σ statistical "tension" evident in independent determinations of the present-day expansion rate of the Universe (e.g., Riess et al. 2016; Freedman 2021; Di Valentino et al. 2021; Abdalla et al. 2022) and, to a lesser extent, the 2-3σ tension in its matter fluctuation spectrum (e.g., Leauthaud et al. 2017; Joudaki et al. 2018; Heymans et al. 2021). The current generation of cosmological surveys, created by facilities such as the *Euclid* satellite (Euclid Collaboration et al. 2024), the Vera Rubin Observatory (Ivezić et al. 2019), the Roman Space Telescope (Eifler et al. 2021) and the Dark Energy Spectroscopic Instrument (DESI, DESI Collaboration et al. 2016), promise a further revolution in statistical precision. However, realising the statistical promise of such cosmological tests places stringent requirements on all parts of the analysis pipeline including the data processing, error budget, cosmological models and likelihood determinations, mo-

Chaves-Montero J., Angulo R. E., Contreras S., 2023, MNRAS, 521, 937
Chen S., et al., 2024, arXiv e-prints, p. arXiv:2407.04795
Contreras S., Chaves-Montero J., Angulo R. E., 2023, MNRAS, 525, 3149
DES Collaboration et al., 2018, Phys. Rev. D, 98, 043526
DES Collaboration et al., 2022, Phys. Rev. D, 105, 023520
DES and KiDS Collaboration et al., 2023, The Open Journal of Astrophysics, 6, 36
DESI Collaboration et al., 2016, arXiv e-prints, p. arXiv:1611.00036
DESI Collaboration et al., 2024, AJ, 168, 58
DeRose J., et al., 2019, arXiv e-prints, p. arXiv:1901.02401
DeRose J., et al., 2022, Phys. Rev. D, 105, 123520
DeRose J., et al., 2023, J. Cosmology Astropart. Phys., 2023, 054
DeRose et al. in prep. 2025, ,
Di Valentino E., et al., 2021, Astroparticle Physics, 131, 102605
Duffy A. R., Schaye J., Kay S. T., Dalla Vecchia C., 2008, MNRAS, 390, L64
Eifler T., et al., 2021, MNRAS, 507, 1746
Elvin-Poole J., et al., 2023, MNRAS, 523, 3649
Emas et al. in prep. 2025, ,
Euclid Collaboration et al., 2024, arXiv e-prints, p. arXiv:2405.13491
Fang X., Eifler T., Krause E., 2020a, MNRAS, 497, 2699
Fang X., Krause E., Eifler T., MacCrann N., 2020b, J. Cosmology Astropart. Phys., 2020, 010
Fosalba P., Gaztañaga E., Castander F. J., Crocce M., 2015, MNRAS, 447, 1319
Freedman W. L., 2021, ApJ, 919, 16
Gatti M., et al., 2021, MNRAS, 504, 4312
Giblin B., et al., 2021, A&A, 645, A105
Guzik J., Seljak U., 2001, MNRAS, 321, 439

Today we'll discuss …



- Where to use references in your work
- How to find the references you need
- Example tools for organizing your references
- How to include references in a LaTeX template

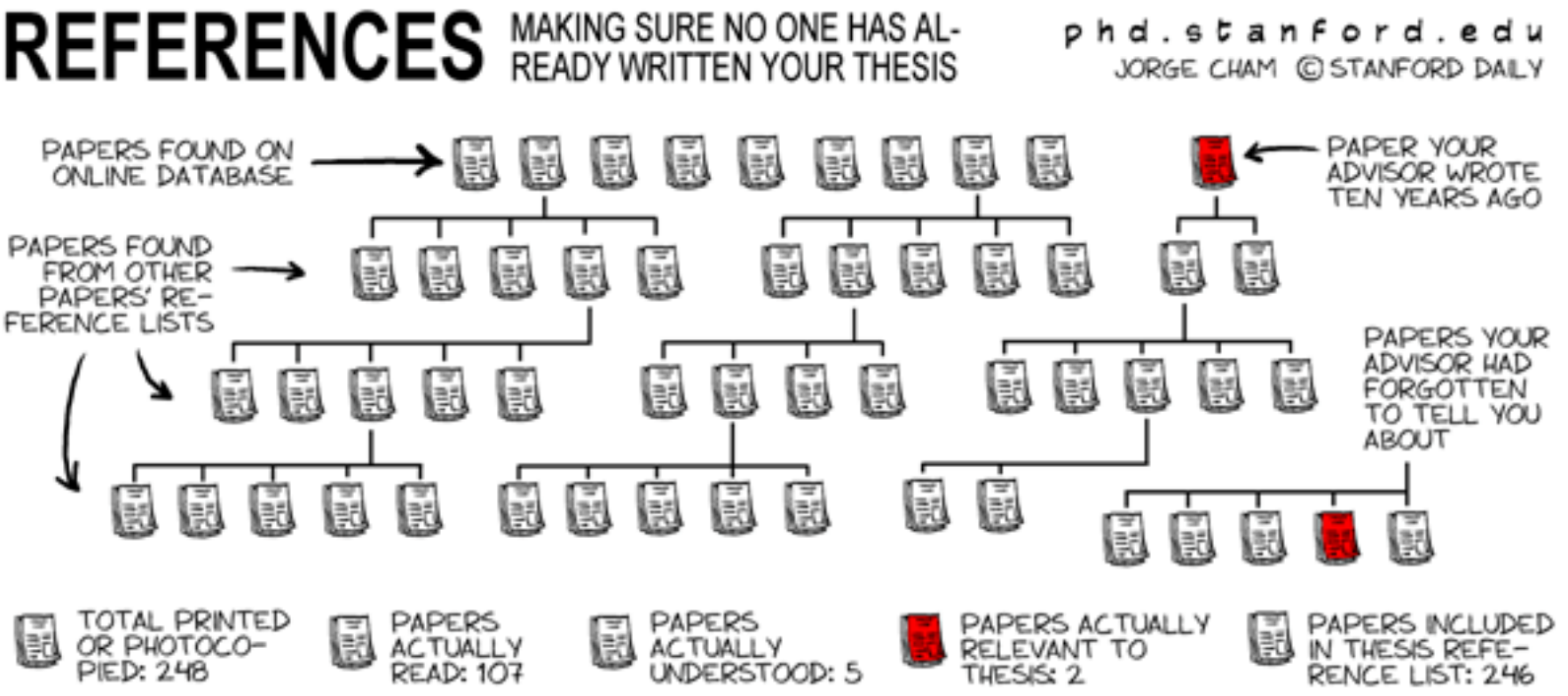
Where to use references in your work

- **Sourcing all technical statements or claims**
  (outside your own methods or very generic statements)
- **Pointing to fuller details** than you want to describe
- For broad statements you can cite **review articles**
- For more technical statements, note useful references as soon as you find them, so you don't lose them!

There is no need to fully understand a paper in order to use it as a reference! *(see next class)*

How to find references

The number of possible references will initially seem infinite, **where do you start from**?



How to find references

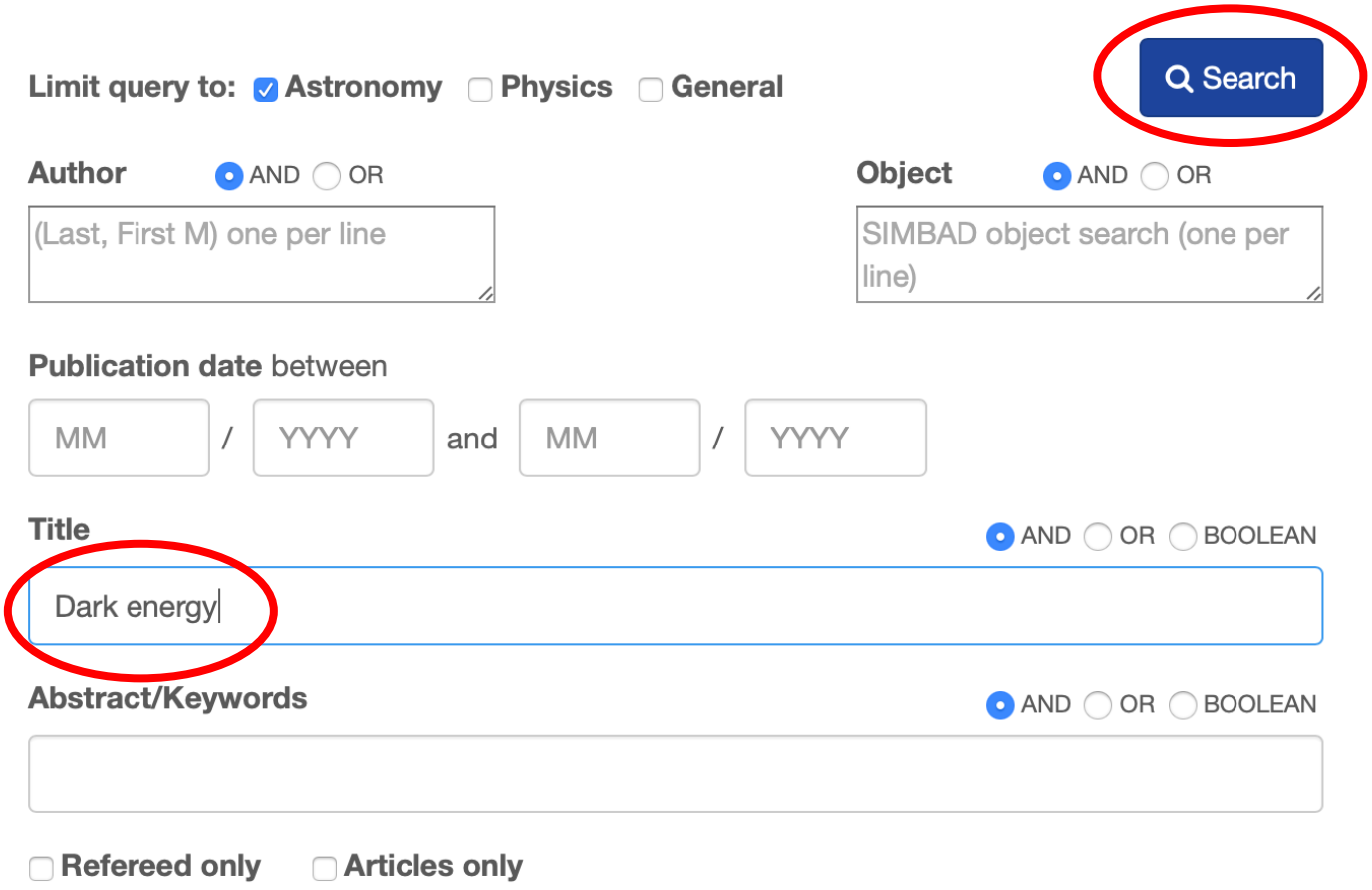
- You can start from **1 or 2 key papers** in your topic
  (… as recommended by your supervisor)
- Use the **introductions of those papers** to find references to related papers (… and those introductions …)
- You can identify the **most highly-cited** papers in your topic by searching on databases in your field
- Be aware of the difference between **primary** and **secondary** references and when to use them
- **Googling the topic** is also a useful check! Be wary of using AI tools to find references though…

Databases of papers

- Widely-used databases in **Astrophysics**:
  - Astrophysics Data System (https://ui.adsabs.harvard.edu)
  - The "ArXiv" (https://arxiv.org/archive/astro-ph) which updates daily with newly-released papers
- Widely-used databases in **Physics** / **Optics** include:
  - American Physical Society (https://www.aps.org)
  - American Institute of Physics (https://ww2.aip.org)
  - Optica publishing group (https://opg.optica.org)
- Databases allow searches by author, title, keyword, etc.

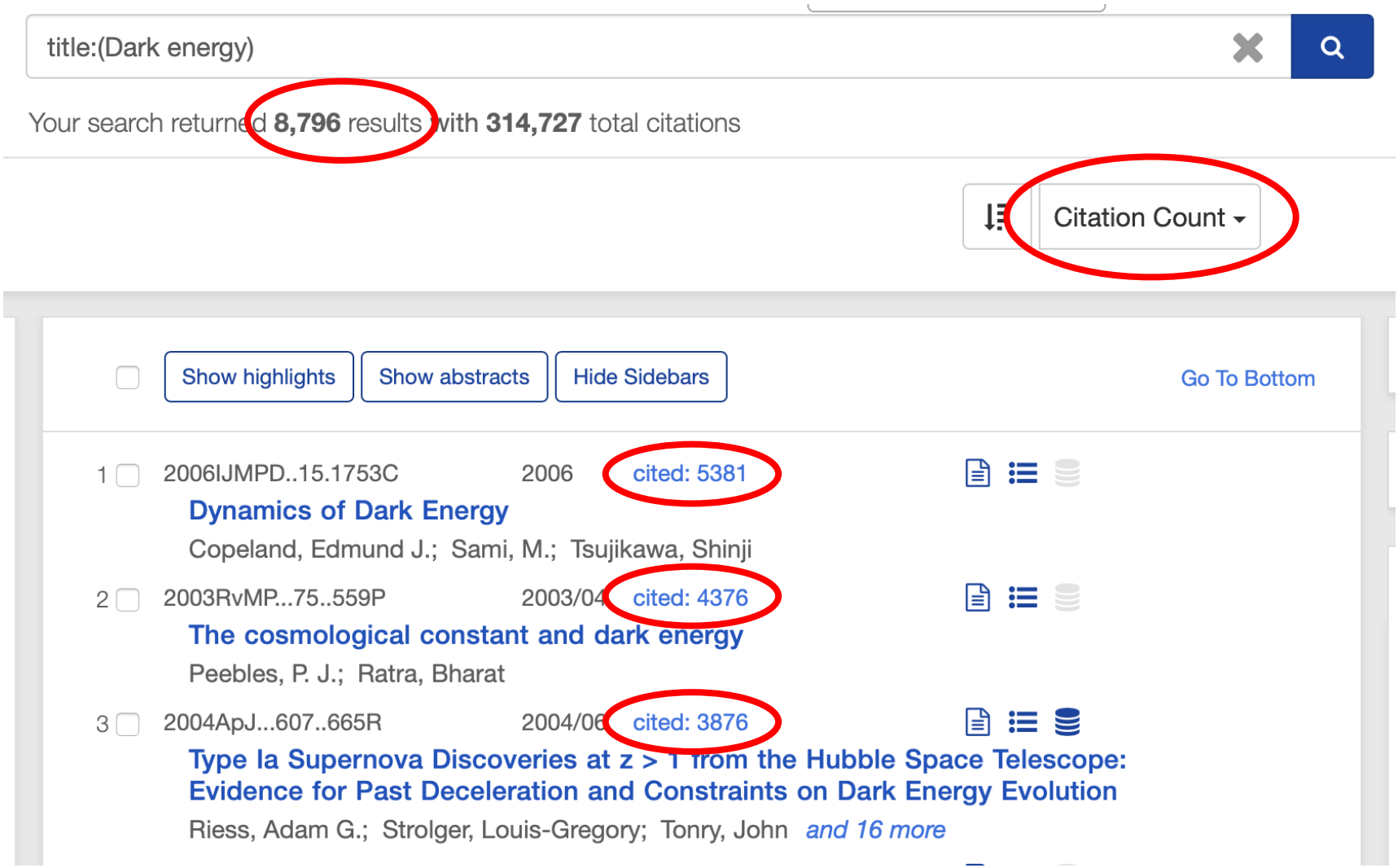
How to find highly-cited references

Astrophysics example using https://ui.adsabs.harvard.edu/ to find highly-cited papers with "dark energy" in the title



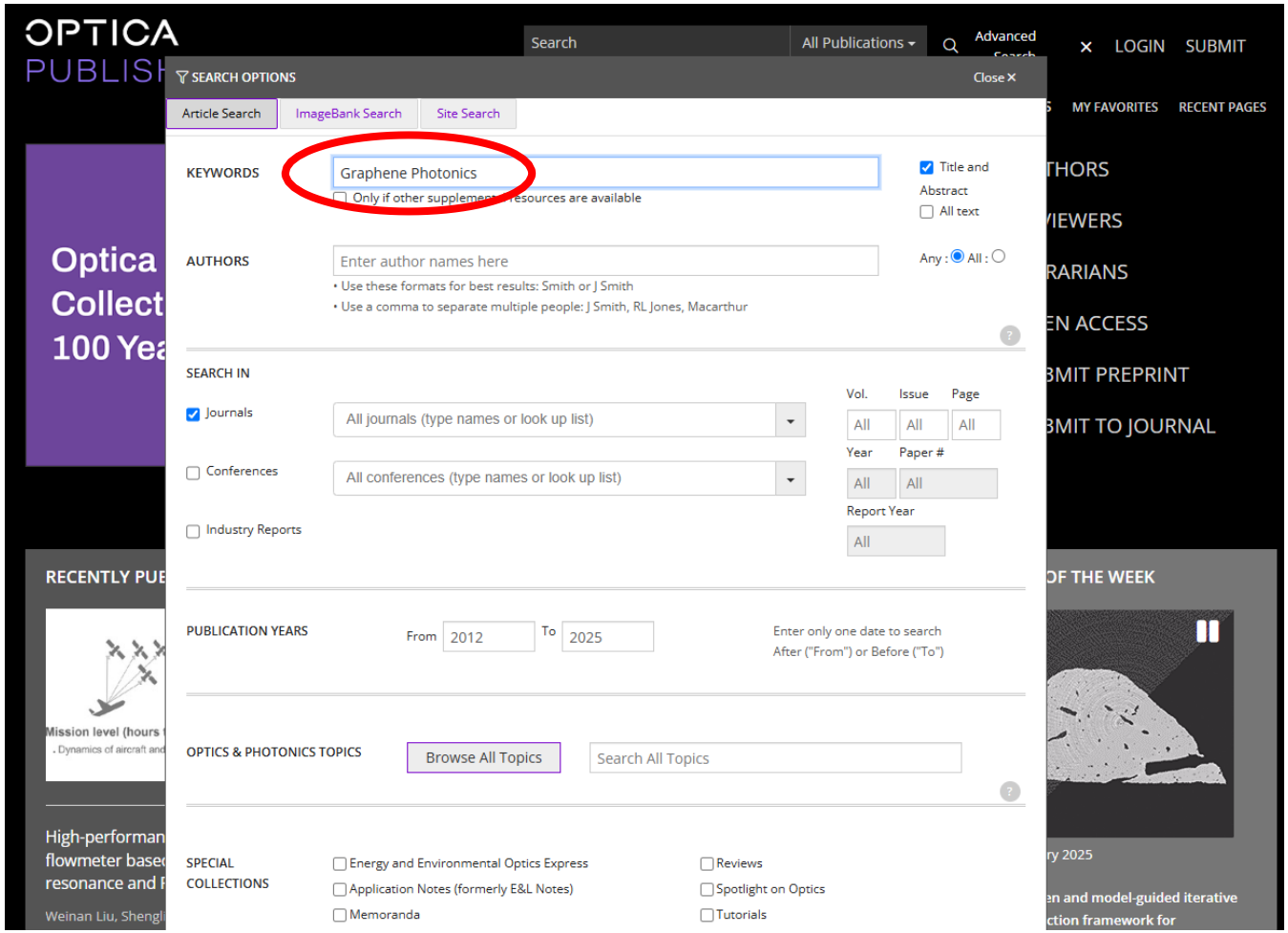
How to find highly-cited references

Astrophysics example using https://ui.adsabs.harvard.edu/



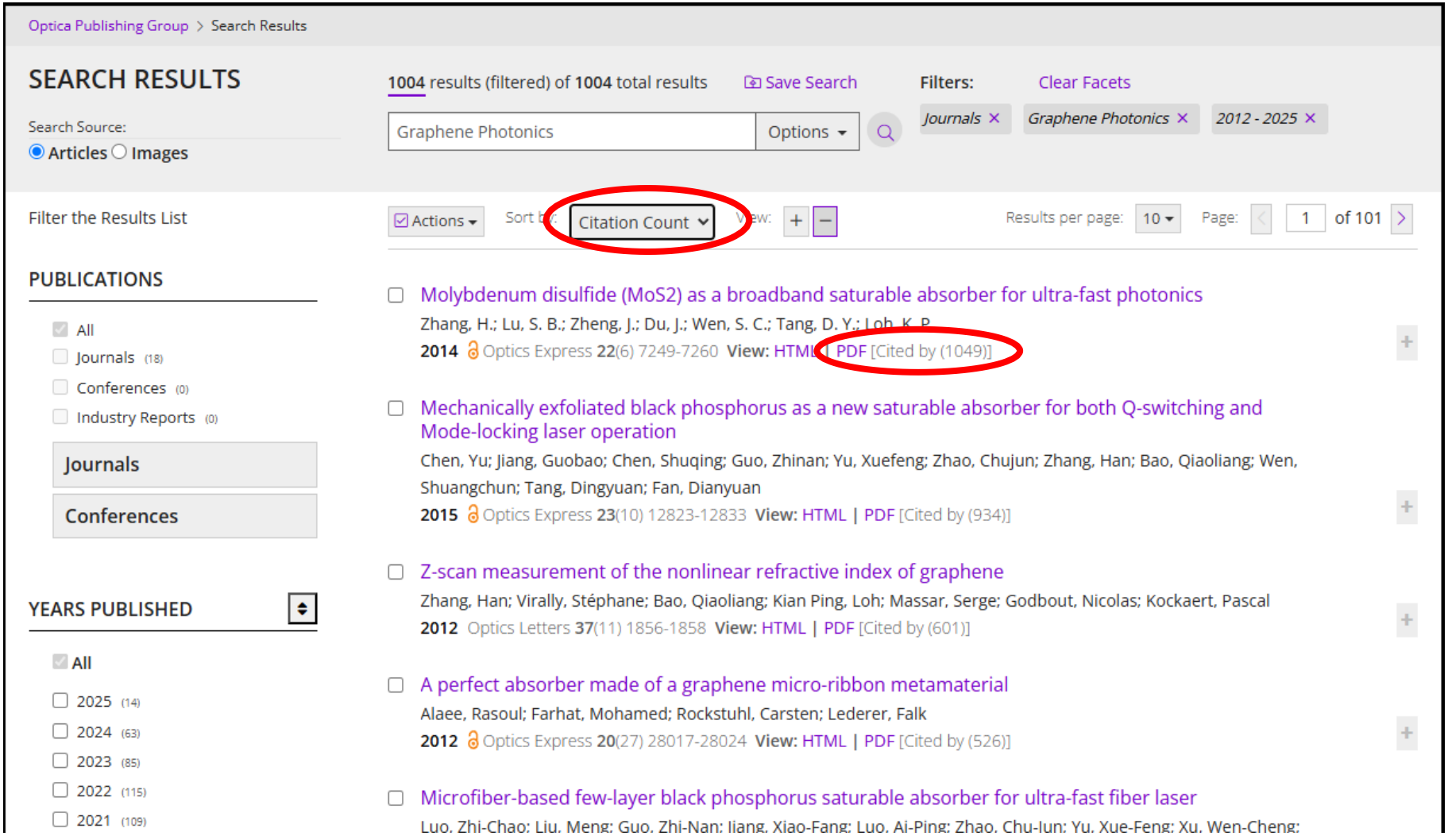
How to find highly-cited references

Optics example using https://opg.optica.org/ to find highly-cited papers with "graphene photonics" in the title



How to find highly-cited references

Optics example using https://opg.optica.org/



Primary and secondary references

A primary reference is the original source, and a secondary reference applies that source

Let's say we want to find a reference for:

"the Kennicutt-Schmidt Law for star formation"



Primary reference:

Kennicutt et al. (1998), ApJ, 498, 451, "The global Schmidt law in star-forming galaxies" (4,604 citations)

Example secondary reference:

Wilson et al. (2019), ApJ, 882, 5, "The Kennicutt-Schmidt Law and Gas Scale Height in Luminous and Ultraluminous Infrared Galaxies" (54 citations)

How to organise references

- You can organize your references using software tools such as **Zotero**
  (free to download from https://www.zotero.org)

- Tag your references with **keywords** that will help you remember their significance later – at first, these can be very simple/broad

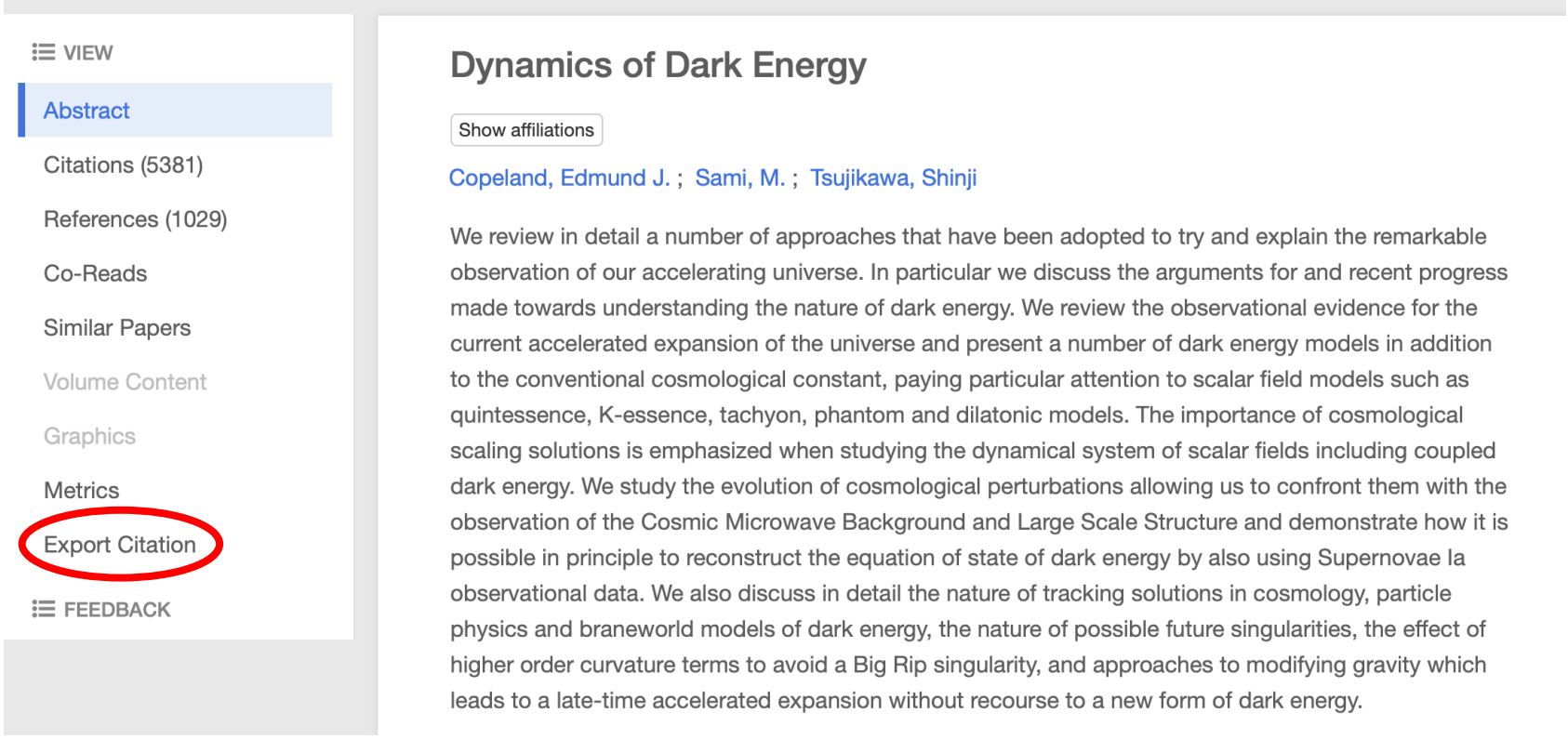
- As you learn more about your topic, you can refine your descriptions of each reference

How to include references

Common method using LaTeX / Overleaf

See Word / EndNote example in the slides

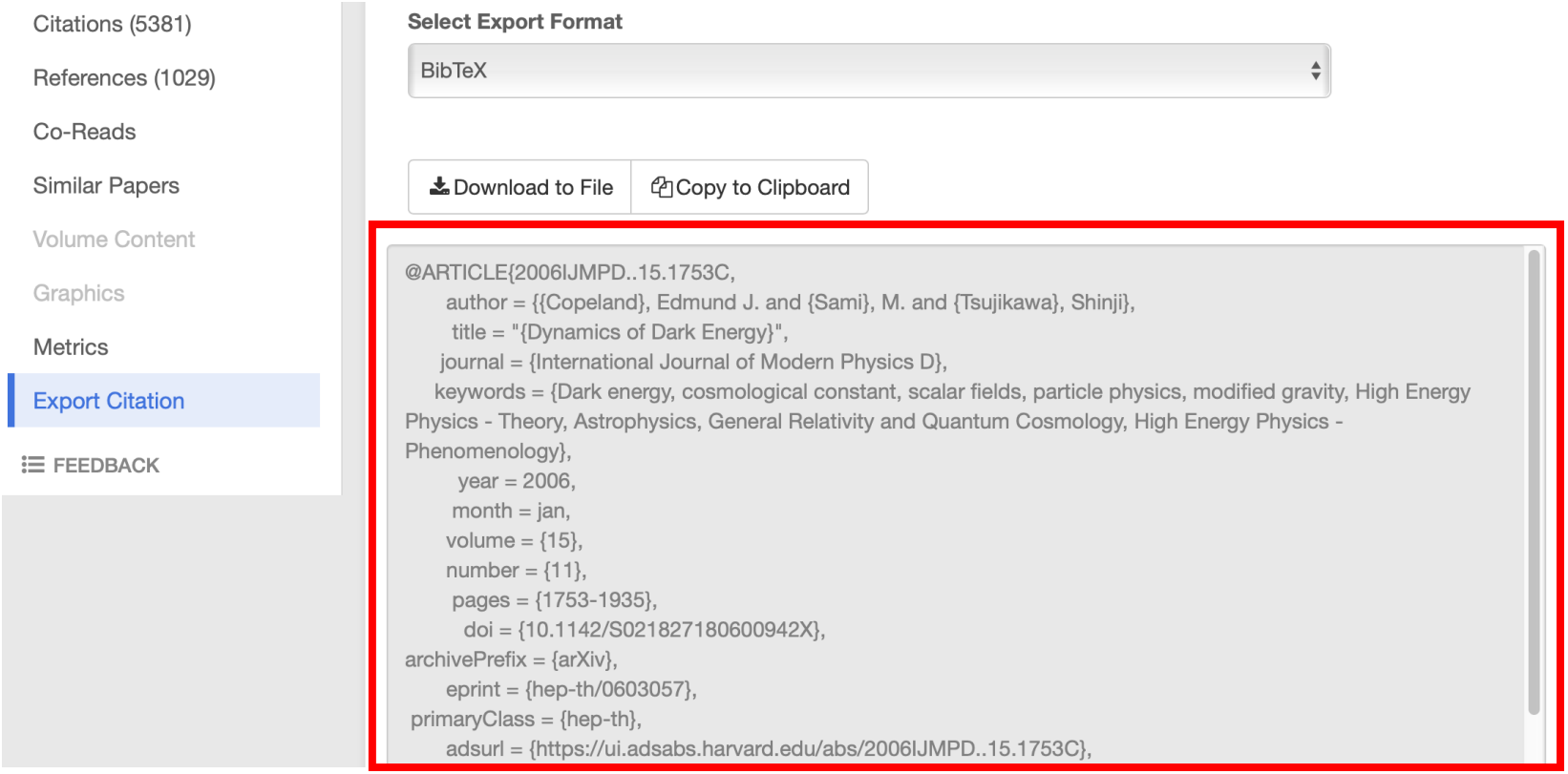
- Create **.bib file** listing your references



How to include references

Common method using LaTeX / Overleaf

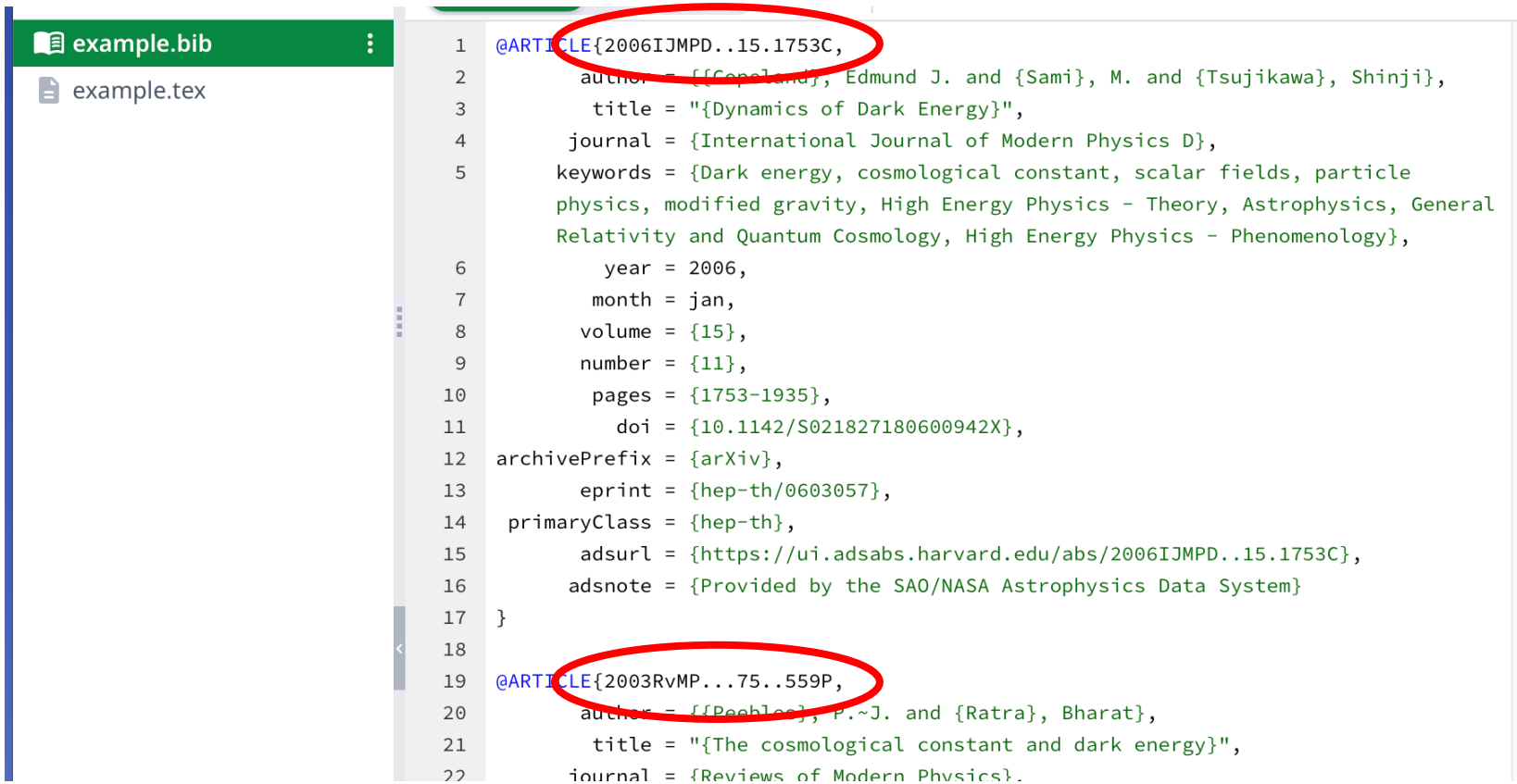
- Create **.bib file** listing your references



How to include references

Common method using LaTeX / Overleaf

- Create **.bib file** listing your references



How to include references

Common method using LaTeX / Overleaf

- Use a latex package such as **natbib**

```
1  \documentclass{article}
2
3  \usepackage[authoryear,round]{natbib}
4  \bibliographystyle{abbrvnat}
5
6  \begin{document}
7
8  \section{Example of references using natbib}
9
10 Here is an example of how to insert references using natbib!
11
12 Examples of papers about dark energy with lots of citations are \cite{2006IJMPD..15.1753C} and
   \cite{2003RvMP...75..559P}.
13
14 People have written highly-cited papers about dark energy \citep{2006IJMPD..15.1753C,
   2003RvMP...75..559P}.
15
16 \bibliography{example}
17
18 \end{document}
```

This means we're using the "Author et al. (Year)" citation style with round brackets

Citing papers in the flow of a sentence

Citing papers after a sentence, in brackets

Bibliography file is called example.bib

How to include references

Common method using LaTeX / Overleaf

- Use a latex package such as **natbib**

Appears like …

1 Example of references using natbib

Here is an example of how to insert references using natbib!

Examples of papers about dark energy with lots of citations are Copeland et al. (2006) and Peebles and Ratra (2003).

People have written highly-cited papers about dark energy (Copeland et al., 2006; Peebles and Ratra, 2003).

References

E. J. Copeland, M. Sami, and S. Tsujikawa. Dynamics of Dark Energy. *International Journal of Modern Physics D*, 15(11):1753–1935, Jan. 2006. doi: 10.1142/S021827180600942X.

P. J. Peebles and B. Ratra. The cosmological constant and dark energy. *Reviews of Modern Physics*, 75(2):559–606, Apr. 2003. doi: 10.1103/RevModPhys.75.559.

Activities

- Try looking through the introduction of your key papers and finding some more references
- Try downloading Zotero (https://www.zotero.org) and organizing your references
- Try making a .bib file and including your references in a LaTeX document (e.g. using natbib) or a Word document (e.g. using EndNote)

That’s all for today!

Example using EndNote

Physical Review Letters

MILESTONE | FREE TO READ

Negative Refraction Makes a Perfect Lens

J. B. Pendry

Show more

Phys. Rev. Lett. **85**, 3966 – **Published 30 October, 2000**

DOI: https://doi.org/10.1103/PhysRevLett.85.3966

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Outline | Inform

- Citing Articles (9990)
- Abstract
- Collections
- Comments & Replies
- Original Article
- References

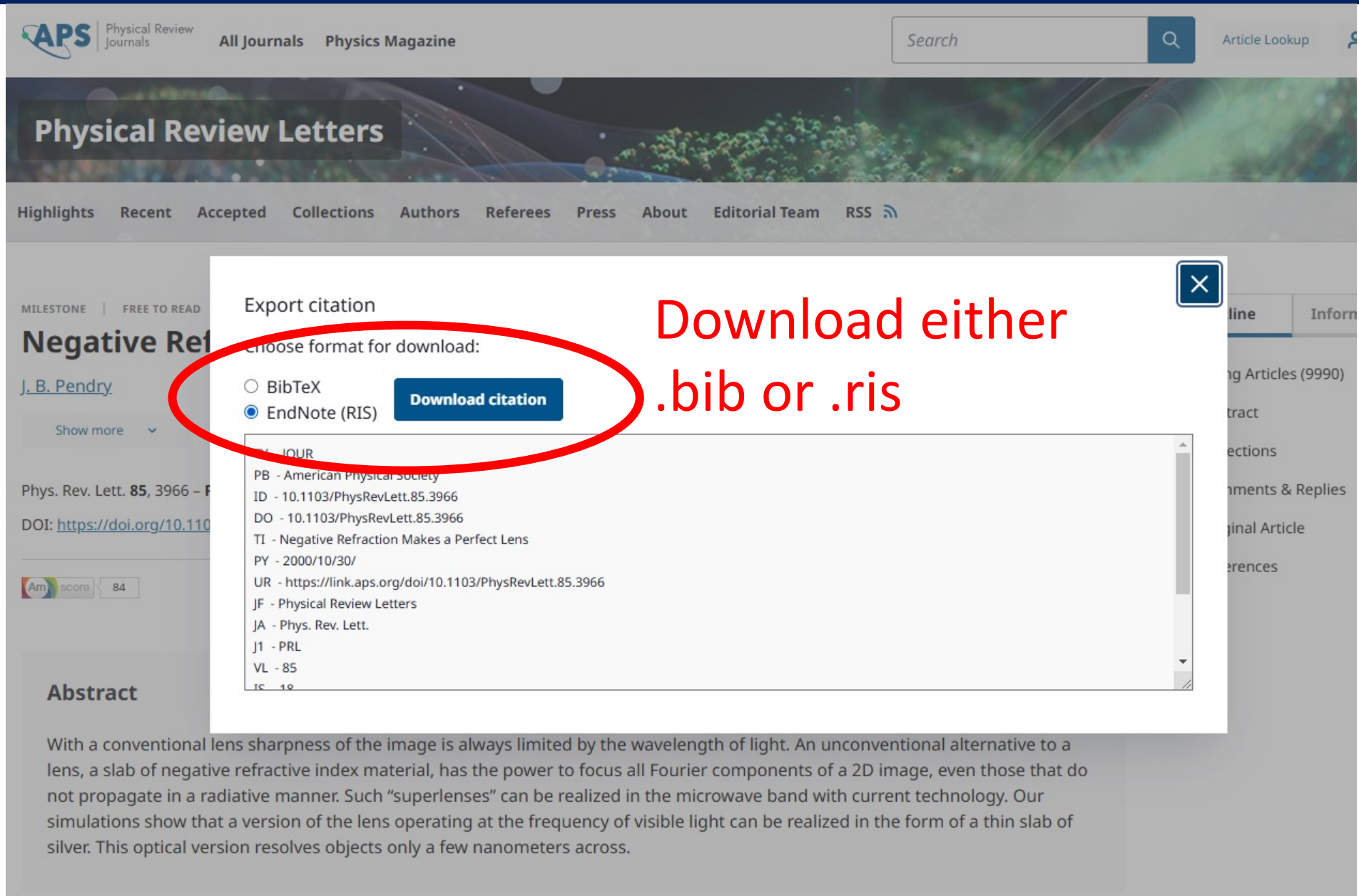
Am score 84 | Citations 10k | Show metrics

Find the paper whose citation you want

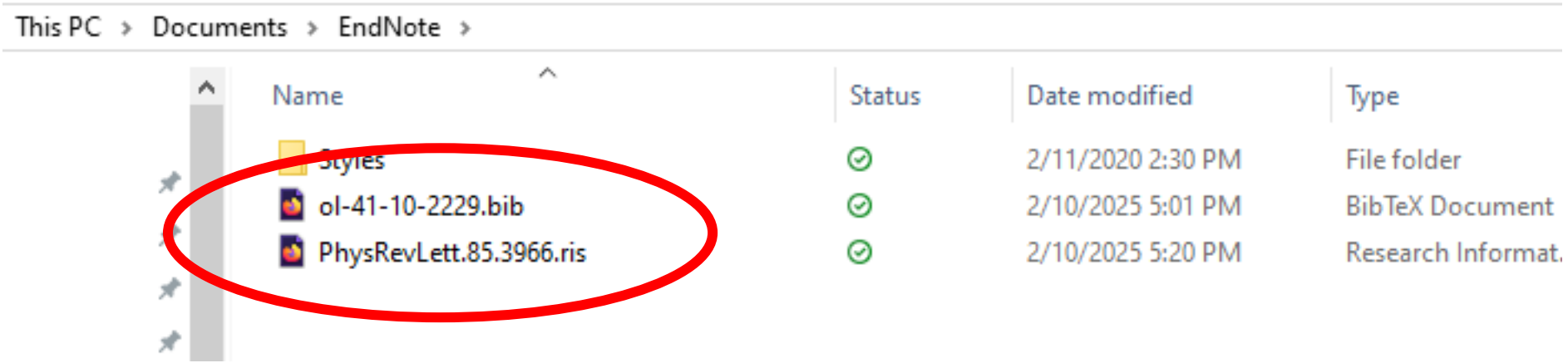
Abstract

With a conventional lens sharpness of the image is always limited by the wavelength of light. An unconventional alternative to a lens, a slab of negative refractive index material, has the power to focus all Fourier components of a 2D image, even those that do not propagate in a radiative manner. Such "superlenses" can be realized in the microwave band with current technology. Our simulations show that a version of the lens operating at the frequency of visible light can be realized in the form of a thin slab of silver. This optical version resolves objects only a few nanometers across.

Example using EndNote

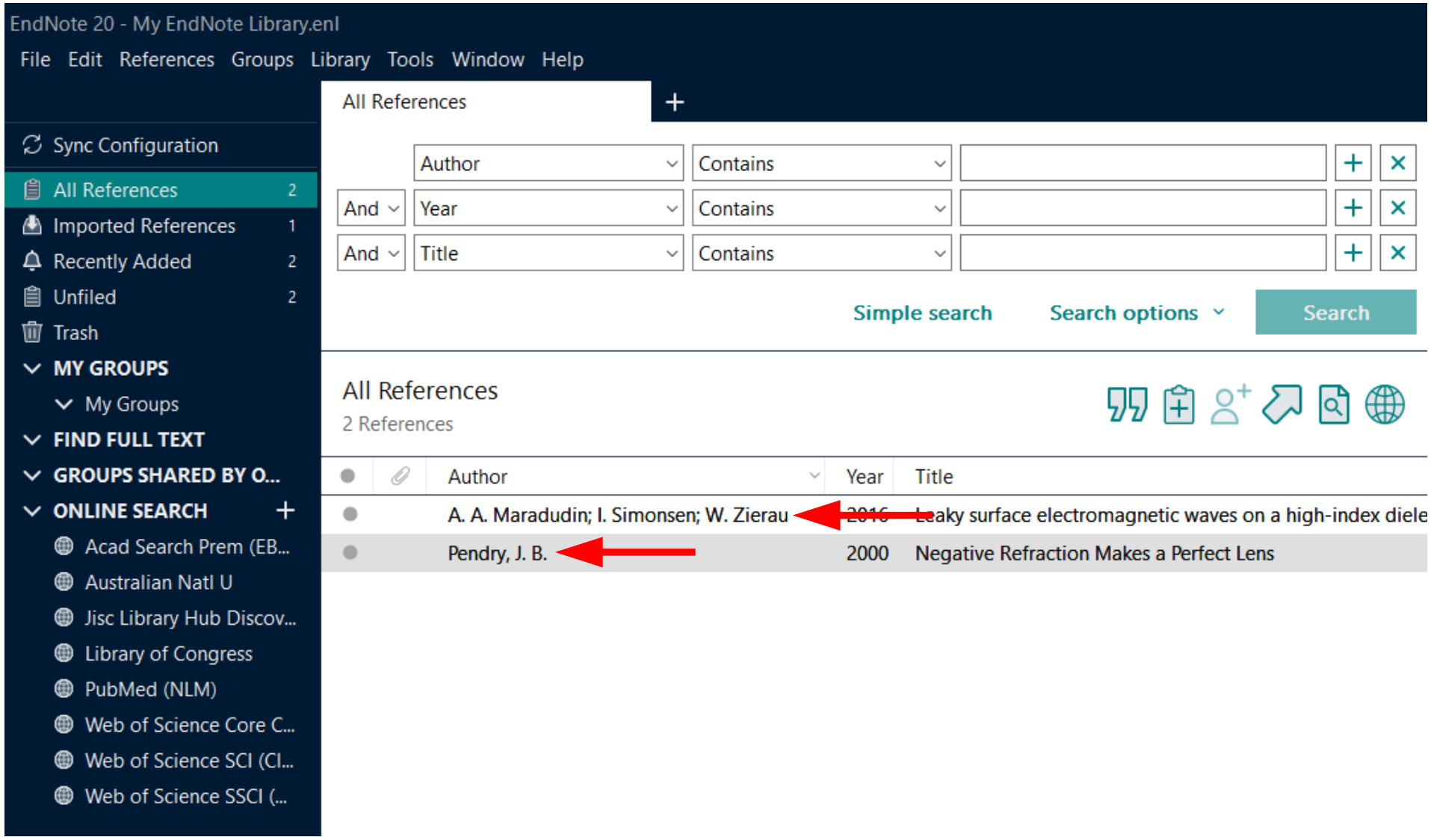


Example using EndNote

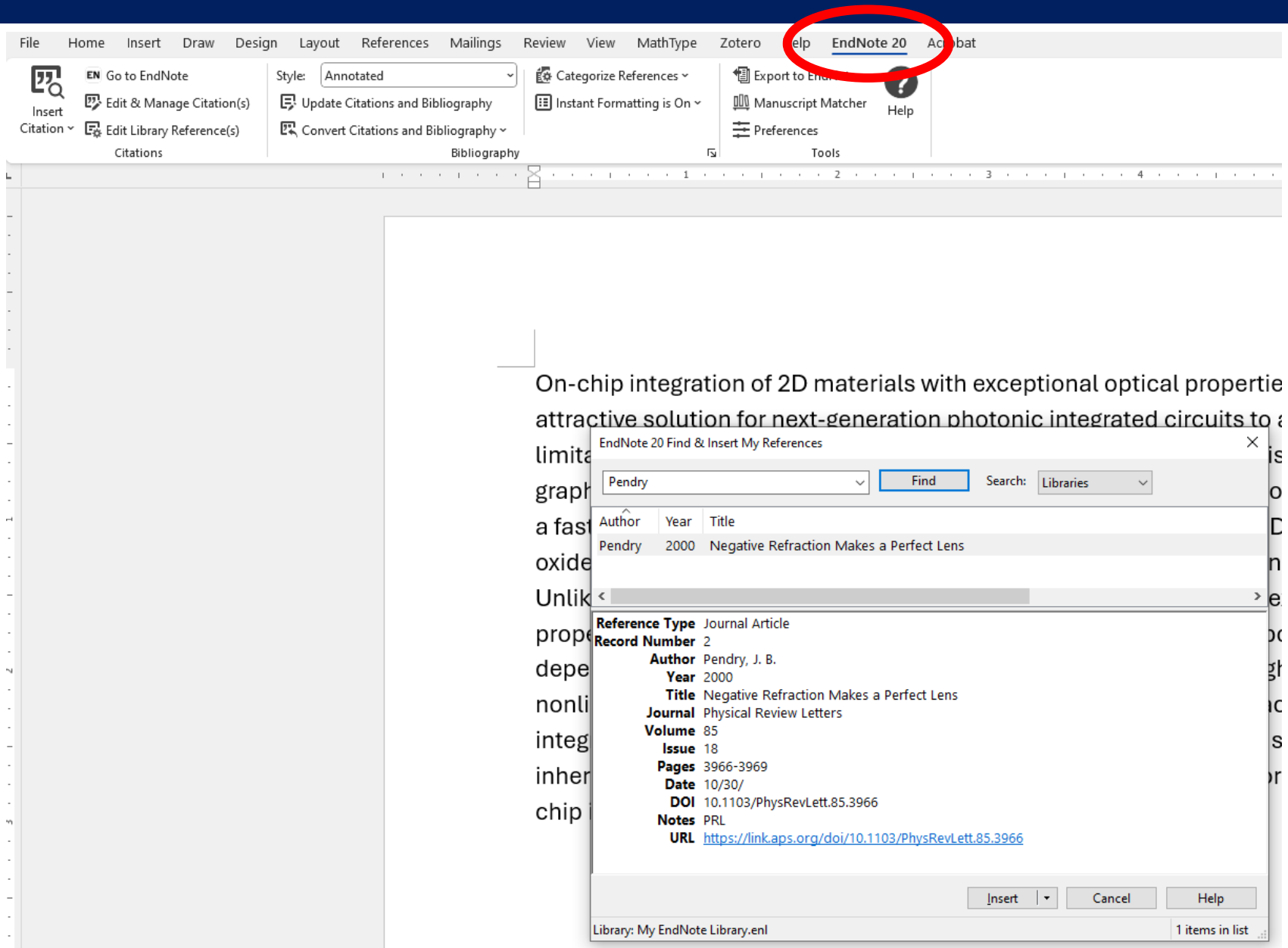


Open the .ris file with Endnote by right clicking, or import .bib using File> Import menu

Example using EndNote



Example using EndNote



Example using EndNote

On-chip integration of 2D materials (Pendry 2000) with exceptional optical properties provides an attractive solution for next-generation photonic integrated circuits to address the limitations of conventional bulk integrated platforms. Since the first isolation of graphene in 2004, atomically thin 2D materials have received enormous interests, with a fast-growing family including transition metal dichalcogenides (TMDCs), graphene oxide (GO), black phosphorus (BP), hexagonal boron nitride (hBN), and many others. Unlike their bulk counterparts, 2D materials demonstrate a range of exceptional properties, such as ultrahigh carrier mobility, broadband optical response, layer-dependent optical bandgaps, significant material anisotropy, and high optical nonlinearity. Moreover, unlike the common lattice mismatch issue faced with bulk integrated materials, 2D materials are feature by dangling-bond free surface and an inherent propensity to adhered through van der Waals. This allows for stress-free on-chip integration and provides a high flexibility in device design.

Pendry, J. B. (2000). "Negative Refraction Makes a Perfect Lens." Physical Review Letters 85(18): 3966-3969.